Biometrika (2005), **92**, 3, *pp.* 647–666
© 2005 Biometrika Trust
Printed in Great Britain

The accelerated gap times model

BY ROBERT L. STRAWDERMAN

Department of Biological Statistics and Computational Biology, Cornell University, Ithaca, New York 14853-7801, U.S.A.

rls54@cornell.edu

SUMMARY

This paper develops a new semiparametric model for the effect of covariates on the conditional intensity of a recurrent event counting process. The model is a transparent extension of the accelerated failure time model for univariate survival data. Estimation of the regression parameter is motivated by semiparametric efficiency considerations, extending the class of weighted log-rank estimating functions originally proposed in Prentice (1978) and subsequently studied in detail by Tsiatis (1990) and Ritov (1990). A novel rank-based one-step estimator for the regression parameter is proposed. An Aalen-type estimator for the baseline intensity function is obtained. Asymptotics are handled with empirical process methods, and finite sample properties are studied via simulation. Finally, the new model is applied to the bladder tumour data of Byar (1980).

Some key words: Censoring; Counting process; Cox regression; Log-rank statistic; Martingale; Multiple events; Rank regression; Recurrent events; Survival data.

1. INTRODUCTION

Recurrent events are prevalent in longitudinal studies in biomedical and public health settings. In these studies, investigators are often interested in the distribution of the time between event occurrences, or 'gap' times, and how this distribution depends on important predictor variables. Existing inference procedures for recurrent event data include both marginal and intensity-based methods. Marginal methods typically forcus on the cumulative rate or mean function, and do not condition on the full event history. Recent examples include Lawless & Nadeau (1995) and Lin et al. (1998, 2000, 2001). Marginal methods are robust to the subject-level correlation structure between events and are useful when interest lies in understanding how population-level characteristics influence the marginal event rate at a fixed point in time. However, strong independent censorship conditions are needed in order to avoid bias (Kalbfleisch & Prentice, 2002, § 9.2).

Intensity models specify how the probability of subsequent recurrence depends on the past event history. Typically less robust to the subject-level correlation structure between events, these models are more useful for studying local process dynamics and predicting recurrence experience at the subject level. Intensity models are also valid under comparatively weak noninformative censoring assumptions. To date, the literature for recurrent event data has focused on extensions of 'Andersen–Gill' intensity models (Andersen & Gill, 1982) and semi-Markov models. Andersen–Gill models extend the Cox regression model (Cox, 1972a) by assuming the underlying event intensity to be the product

of an unknown deterministic baseline intensity function and a known predictable function of the past event history. The prevalent semi-Markov regression models are derived from the 'modulated renewal process' of Cox (1972b), parameterising the baseline intensity in terms of the sojourn time, or time elapsed since the most recent event. Specific examples include Prentice et al. (1981), Oakes (1992), Chang & Hsiung (1994), Oakes & Cui (1994) and Chang & Wang (1999). Nonparametric estimation in related settings has been considered by Gill (1981), Wang & Chang (1999) and Peña et al. (2001).

This paper proposes a new semiparametric semi-Markov intensity model for recurrent event data. Similarly to the accelerated failure time model, covariates serve to accelerate or decelerate the gap times between events, providing the model with a 'quite direct physical interpretation', in the words of D. R. Cox (Reid, 1994, p. 450). Estimation and inference procedures for this 'accelerated gap times' model are considered for the case of time-independent, or 'baseline', covariates.

2. THE MODEL

2·1. *Motivation and definition*

Consider first a single subject with a p-dimensional covariate vector Z experiencing repeated events at times $0 < U_1 < U_2 < \ldots$. Define the jth gap time as $T_j = U_j - U_{j-1}$, where $j \geqslant 1$ and $U_0 = 0$. Let $V_1, V_2, \ldots$ be independent and identically distributed continuous random variables with cumulative distribution function F_0. Conditionally upon Z, suppose that $T_1, T_2, \ldots$ are independent random variables, where $T_j = V_j \exp(-\theta_0' Z)$ for each $j \geqslant 1$. Observe that Z accelerates or decelerates the 'baseline' gap times V_j $(j \geqslant 1)$ similarly to the accelerated failure time model.

Evidently, the hazard function of T_j given Z is $\lambda_0(te^{\theta_0' Z})e^{\theta_0' Z}$, where $\lambda_0(.)$ is the hazard function associated with $F_0(.)$. Define $U_m = \sum_{j=1}^m T_j$, and let $N^v(t) = \max(m: U_m \leqslant t)$ for $t \geqslant 0$. Then, in the absence of censoring and given Z, $N^v(t)$ is a renewal process with cumulative intensity

$$\Lambda(t|Z) = \int_0^t \lambda_0\{e^{\theta_0' Z} R^v(s)\} e^{\theta_0' Z} ds, \tag{1}$$

where $R^v(s) = s - U_{N^v(s-)}$ (Daley & Vere-Jones, 1988, p. 523).

Suppose now that there are n independent subjects, the ith being observed over $[0, \tau_i]$. Let the observed data be $\{N_i^\dagger(u), Y_i^\dagger(u), Z_i, u \geqslant 0\}$, for $i = 1, \ldots, n$, where $N_i^\dagger(u) = N_i^v(u \wedge \tau_i)$, $Y_i^\dagger(u) = I(\tau_i \geqslant u)$, and $a \wedge b = \min(a, b)$. It is assumed throughout the paper that Conditions A1–A5 of the Appendix hold. Let $\mathscr{G}_t$ denote the event and covariate histories of all subjects to time t. Lemma 1 now encapsulates the major model assumptions.

LEMMA 1. *For $i = 1, \ldots, n$ define $R_i(u) = u - U_{iN_i^\dagger(u-)}$ and set*

$$A_i^\dagger(t, \theta) = \int_0^t \lambda_0\{e^{\theta' Z_i} R_i(u)\} e^{\theta' Z_i} Y_i^\dagger(u) du.$$

Then $M_i^\dagger(.) = N_i^\dagger(.) - A_i^\dagger(., \theta_0)$, for $i \geqslant 1$, are orthogonal square-integrable zero-mean martingales with respect to $\mathscr{G} = \{\mathscr{G}_t, t \geqslant 0\}$.

2·2. *Related models*

The modulated renewal process model of Cox (1972b) assumes that the cumulative intensity function of $N^v(t)$ equals $\int_0^t \lambda_0\{R^v(s)\} e^{\theta_0' Z} ds$; see also Oakes & Cui (1994). Related

models include those in Prentice et al. (1981), Oakes (1992), Chang & Hsiung (1994) and Chang & Wang (1999). The modulated renewal process and accelerated gap times models are equivalent to the Andersen–Gill model if $\lambda_0(u) = \lambda_0$ for $u \geqslant 0$; otherwise, all three models are distinct.

Lin et al. (1998) propose the marginal model $E\{N^v(t)|Z\} = \mu_0(te^{\beta_0'Z})$, where $\mu_0(.)$ is an increasing function. This 'accelerated means' model neither assumes the gap times T_j are independent nor that they have the same conditional distributions, i.e. given Z. However, there is an interesting correspondence between this model and the accelerated gap times model in the case where both are valid. In particular, if we assume that the accelerated gap times model holds, $E\{N^v(t)|Z\} = \gamma_0(te^{\theta_0'Z})$, where $\gamma_0(t)$ is the renewal function associated with $F_0(.)$. Hence, under the accelerated gap times model, $\mu_0(.) = \gamma_0(.)$ and the interpretation of θ_0 and β_0 coincide. The estimating function for β_0 proposed in Lin et al. (1998) can thus be used for estimating θ_0, provided censoring is independent of the underlying event process given Z. The estimating function (13) derived below is motivated by efficiency considerations and yields a consistent estimator under weaker censoring assumptions. The accelerated gap times model is also a more natural choice if the focus of the analysis is on the recurrence time distribution. For example, with T denoting a generic gap time, it is easy to estimate

$$\mathrm{pr}(T > t|Z) = \exp\{-\Lambda_0(te^{\theta_0'Z})\} \tag{2}$$

and, as a result, obtain median recurrence times and related quantities.

Huang (2002b) proposes the multi-state accelerated sojourn times model, a marginal regression model for the sojourn times between visits to the states of some stochastic process. The model assumes that the state space of the underlying process consists of $K < \infty$ states and that these states are visited in the order $1, \ldots, K$ at random times $U_1 < \ldots < U_K$. For recurrent event data, U_k corresponds to the calendar time of the kth recurrent event. For a single subject, Huang's model assumes that $\log T_j = \beta_j'Z + \varepsilon_j$ $(j = 1, \ldots, K)$, where $T_j = U_j - U_{j-1}$ and ε_j $(j = 1, \ldots, K)$ have some unspecified joint distribution. The model shares an obvious structural similarity to the accelerated gap times model and, similarly to Lin et al. (1998), inference is robust to misspecification of the gap time dependence structure at the expense of strong random censorship assumptions. In fact, Huang imposes the unnecessarily strong assumption that censorship be independent of both covariates and the recurrent event process. Extending his estimation procedures to allow for covariate-dependent censoring requires one to model the censoring mechanism, reducing robustness. In addition, the finite state space assumption implies that each subject can experience at most K events, with an adverse impact on efficiency. The accelerated sojourn times model thus presents a less attractive approach to the analysis of recurrent event data of the same type, particularly when the numbers of events vary substantially across subjects.

3. Semiparametric estimation

3·1. *Introduction*

Prentice (1978) suggested estimating the regression coefficient under the semiparametric accelerated failure time model by inverting a class of weighted linear rank statistics. Tsiatis (1990) subsequently established the asymptotic properties of this class of estimators. Ritov (1990) studied this same problem from a different perspective, establishing a direct correspondence between the estimating functions considered by Prentice and Tsiatis and

those based on semiparametric efficiency considerations for censored linear regression models. In § 3·2, semiparametric efficiency considerations are used to derive a weighted log-rank-type estimating function for θ_0 and also an estimator for $\Lambda_0(.)$ under the accelerated gap times model. The resulting estimating function for θ_0 directly generalises the estimating functions for the accelerated failure time model to the case where each subject experiences serial events of the same type.

3·2. *Information bounds for estimating θ_0 and Λ_0*

To motivate estimators for θ_0 and Λ_0, we first consider the problem of semiparametric efficient estimation. Doing so will require that we be able to move freely between 'calendar time' and 'gap time' stochastic integral representations of the same quantity (Peña et al., 2001). Lemmas 2 and 3 provide the essential tools. Lemma 2 generalises Proposition 1 of Peña et al. (2001). Lemma 3 is a consequence of Lemmas 1 and 2 and well-known properties of martingale integrals. Proofs are therefore omitted. Hereafter, θ denotes some vector in $\mathbb{R}^p$.

LEMMA 2. *For $s, t \geqslant 0$, $i = 1, \ldots, n$ and any bounded integrable $h(.)$,*

$$\int_0^s h\{R_i(v)e^{\theta' Z_i}\}\lambda_0\{R_i(v)e^{\theta' Z_i}\}Y_i^\dagger(v)e^{\theta' Z_i}dv = \int_0^\infty h(r)\lambda_0(r)Y_i(s, r|\theta)dr,$$

where $Y_i(s, t|\theta) = \sum_{j=1}^{N_i^\dagger(s-)} I(T_{ij}e^{\theta' Z_i} \geqslant t) + I[e^{\theta' Z_i}\{\tau_i \wedge s - U_{iN_i^\dagger(s-)}\} \geqslant t]$.

To see the utility of Lemma 2, consider $h(r) = I(r \leqslant t)$ for a fixed $t > 0$. Note that

$$N_i(s, t|\theta) = \int_0^s h\{R_i(v)e^{\theta' Z_i}\}N_i^\dagger(dv) \equiv \sum_{j=1}^{N_i^\dagger(s-)} I(T_{ij}e^{\theta' Z_i} \leqslant t). \tag{3}$$

For fixed $s > 0$, $N_i(s, t|\theta)$ and $Y_i(s, u|\theta)$ essentially track events and 'at risk' status on the baseline gap time scale. Hence, one might expect that the centred process,

$$M_i(s, t|\theta) = N_i(s, t|\theta) - \int_0^t Y_i(s, u|\theta)\lambda_0(u)du, \tag{4}$$

has mean zero at $\theta = \theta_0$. Indeed, using (3), observe that

$$M_i(s, t|\theta_0) = \int_0^s h\{R_i(v)e^{\theta' Z_i}\}M_i^\dagger(dv)$$

by Lemmas 1 and 2 and is thus a zero-mean martingale in $s \geqslant 0$ for each $t \geqslant 0$. Importantly, $M_i(s, t|\theta_0)$ is not a martingale in $t \geqslant 0$ for fixed $s > 0$ (Peña et al., 2001). Lemma 3 shows that $M_i(s, t|\theta_0)$ still exhibits martingale-like behaviour in mean and variance calculations.

LEMMA 3. *Let $N_i(t|\theta) = N_i(\infty, t|\theta)$, $Y_i(t|\theta) = Y_i(\infty, t|\theta)$ and $M_i(t|\theta) = M_i(\infty, t|\theta)$. Suppose that $H_i(.)$ is a $\mathscr{G}$-predictable process for $\mathscr{G}$ defined in Lemma 1. Then, for any $t \geqslant 0$,*

$$\int_0^t H_i(u)M_i(du|\theta) \equiv \int_0^\infty I\{R_i(v)e^{\theta' Z_i} \leqslant t\}H_i\{R_i(v)e^{\theta' Z_i}\}A_i^\dagger(dv, \theta),$$

$$E\left\{\int_0^t H_i(u)M_i(du|\theta_0)\right\} = 0, \quad \operatorname{var}\left\{\int_0^t H_i(u)M_i(du|\theta_0)\right\} = \int_0^t E\{H_i^2(u)Y_i(u|\theta_0)\}\lambda_0(u)du.$$

We now turn to the problem of efficient estimation. Suppose first that $\Lambda_0(.)$ is known. Then Jacod's point-process loglikelihood (Andersen et al., 1993, Cor. II.7.3) may be used to construct a score function for θ. To be specific, for $\theta = \theta_0$, we find that

$$S_Q(\theta_0) = \sum_{i=1}^{n} \int_0^\infty Z_i Q\{R_i(u) e^{\theta_0' Z_i}\} M_i^\dagger(du), \tag{5}$$

where $M_i^\dagger(.)$ is defined as in Lemma 1 and hereafter $Q(u) = 1 + u\{\dot{\lambda}_0(u)/\lambda_0(u)\}$, with $\dot{\lambda}_0(u)$ denoting the derivative of $\lambda_0(u)$. Now suppose that θ_0 is known and consider the parameterised family of submodels $(\theta_0, \Lambda_0) \to (\theta_0, \Lambda_\alpha)$, where $\Lambda_\alpha(t) = \int_0^t \lambda_0(u)\{1 + \frac{1}{2}\alpha\eta(u)\}^2 du$, for $\alpha \in \mathbb{R}$, and $\eta(.)$ is some bounded element of a suitable Hilbert space $\mathbb{H}$. Note that $(\theta_0, \Lambda_\alpha)$ reduces to (θ_0, Λ_0) at $\alpha = 0$. A score function for $\Lambda_0(.)$ (van der Vaart, 1998, § 25.5.1) can be derived from the same loglikelihood under this family of parametric submodels if we differentiate it with respect to α and set $\alpha = 0$: for a given $\eta(.)$ we obtain

$$B_{\theta_0,\Lambda_0}(\eta) = \sum_{i=1}^{n} \int_0^\infty \eta\{R_i(u) e^{\theta_0' Z_i}\} M_i^\dagger(du). \tag{6}$$

Informally, the infinite set of score functions obtained by letting $\eta(.)$ vary over $\mathbb{H}$ generates the tangent space for $\Lambda_0(.)$.

The efficient score for θ_0 is obtained by subtracting from $S_Q(\theta_0)$ its orthogonal projection on to the tangent space for $\Lambda_0(.)$ (van der Vaart, 1998, § 25.4). To be more precise we first determine $\varphi_k(.) \in \mathbb{H}$, for $k = 1, \ldots, p$, such that $E[\{S_{Qk}(\theta_0) - B_{\theta_0,\Lambda_0}(\varphi_k)\} B_{\theta_0,\Lambda_0}(\eta)] = 0$ for all $\eta \in \mathbb{H}$, where $S_{Qk}(\theta_0)$ is the kth element of the p-vector $S_Q(\theta_0)$, and then we compute $S_{\text{eff}}(\theta_0) = S_Q(\theta_0) - \{B_{\theta_0,\Lambda_0}(\varphi_1), \ldots, B_{\theta_0,\Lambda_0}(\varphi_p)\}'$.

Define $s^{(j)}(u|\theta) = E\{Z_1^{\otimes j} Y_1(u|\theta)\}$, for $j = 0, 1, 2$, where $v^{\otimes k}$ $(k = 0, 1, 2)$ respectively denote $1^{p\times 1}$, v and vv' for a $p \times 1$ vector v. Also, let $\varphi(u) = \{\varphi_1(u), \ldots, \varphi_p(u)\}'$. Proposition 1 summarises the results; proofs of propositions and theorems are given in the Appendix.

PROPOSITION 1. *Suppose that* $\int_0^\infty Q^2(u)\lambda_0(u)du < \infty$. *Then* $\varphi(u) = Q(u)\mathscr{E}(u)$, *where* $\mathscr{E}(u) = \mathscr{E}(u|\theta_0)$ *and* $\mathscr{E}(u|\theta) = s^{(1)}(u|\theta)/s^{(0)}(u|\theta)$. *The efficient score for* θ_0 *is* $S_{\text{eff}}(\theta_0)$, *where*

$$S_{\text{eff}}(\theta) = \sum_{i=1}^{n} \int_0^\infty Q(u)\{Z_i - \mathscr{E}(u|\theta)\} M_i(du|\theta). \tag{7}$$

Assume that $\Sigma_Q = n^{-1} E\{S_{\text{eff}}(\theta_0) S_{\text{eff}}'(\theta_0)\} > 0$, *where*

$$\Sigma_Q = \int_0^\infty E[\{Z_1 - \mathscr{E}(u)\}^{\otimes 2} Y_1(u|\theta_0)] Q^2(u)\lambda_0(u)du. \tag{8}$$

Then the semiparametric information bound for estimating θ_0 *is* Σ_Q^{-1}.

The score for $\Lambda_0(.)$ is given in (6). Using Lemma 2, one may write this as

$$B_{\theta_0,\Lambda_0}(\eta) = \sum_{i=1}^{n} \int_0^\infty \eta(u)\{N_i(du|\theta_0) - Y_i(u|\theta_0)\Lambda_0(du)\},$$

where $\eta(.) \in \mathbb{H}$ is bounded. Define the Aalen-type estimator

$$\hat{\Lambda}_0(t|\theta) = \int_0^t \frac{\sum_{i=1}^n N_i(du|\theta)}{\sum_{i=1}^n Y_i(u|\theta)}. \tag{9}$$

Evidently, $B_{\theta_0,\hat{\Lambda}_0(.|\theta_0)}(\eta)=0$ for any $\eta \in \mathbb{H}$, showing (9) to be the solution to a score equation. The results of Peña et al. (2001) imply that $\hat{\Lambda}_0(.|\theta_0)$ is an efficient estimator of $\Lambda_0(.)$. However, since θ_0 is unknown, a consistent estimator $\hat{\theta}$ must be used in place of θ_0, and $\hat{\Lambda}_0(t|\hat{\theta})$ is no longer efficient in general. Proposition 2 provides the relevant information bound.

PROPOSITION 2. *Suppose that* $s^{(0)}(u|\theta_0)>0$ *for* $u \in [0, t]$ *and define*

$$A_Q(t)=\int_0^t \frac{s^{(1)}(u|\theta_0)}{s^{(0)}(u|\theta_0)} Q(u)\lambda_0(u)du. \tag{10}$$

Then, under the conditions of Proposition 1,

$$\int_0^t \frac{\lambda_0(u)}{s^{(0)}(u|\theta_0)} du + A_Q'(t)\Sigma_Q^{-1} A_Q(t) \tag{11}$$

is the semiparametric information bound for $\Lambda_0(t)$.

3·3. *Estimation of* θ_0 *and* Λ_0

Using (9), we can estimate $\Lambda_0(t)$ by $\hat{\Lambda}_0(t|\hat{\theta})$, where $\hat{\theta}$ is some consistent estimator of θ_0. Whether or not $\hat{\Lambda}_0(t|\hat{\theta})$ is efficient depends primarily on the choice of $\hat{\theta}$; see Theorem 4 below. The main challenge therefore lies in the estimation of θ_0.

To use (7) for estimating θ_0, we must estimate both $\mathscr{E}(.|\theta)$ and $Q(.)$. Estimation of $\mathscr{E}(.|\theta)$ is straightforward: with

$$\mathbb{S}^{(k)}(t|\theta)=n^{-1}\sum_{j=1}^{n} Z_j^{\otimes k} Y_j(t|\theta), \tag{12}$$

we can use $\mathbb{E}(t|\theta)=\mathbb{S}^{(1)}(t|\theta)/\mathbb{S}^{(0)}(t|\theta)$. Nonparametric estimation of $Q(.)$ is more difficult. Given an estimator $\hat{\Lambda}_0(t|\hat{\theta})$, estimation of $Q(.)$ using kernel methods is a possibility. However, such estimators are known to suffer from bias, are sensitive to bandwidth selection, and exhibit slow rates of convergence (Scott, 1992, §§ 6.1–6.2). In general, it is difficult to estimate $Q(.)$ well with sample sizes that typically arise in applications.

For estimation under the accelerated failure time model, Tsiatis (1990) and Ritov (1990) propose replacing the unknown optimal weight with a computable, data-dependent weight $W(u|\theta)$. If we substitute $\mathbb{E}(.|\theta)$ for $\mathscr{E}(.|\theta)$ and $W(.|\theta)$ for $Q(.)$ in (7), algebra leads to the weighted class of estimating functions

$$\bar{S}_W(\theta)=\frac{1}{n}\sum_{i=1}^{n}\int_0^\infty W(u|\theta)\{Z_i-\mathbb{E}(u|\theta)\}N_i(du|\theta). \tag{13}$$

With $N_i=N_i^\dagger(\tau_i-)$ denoting the observed number of events for subject i, define $\tilde{X}_{ij}(\theta)=T_{ij}e^{\theta' Z_i}$ for $j \leqslant N_i$ and $\tilde{X}_{ij}(\theta)=(\tau_i-U_{iN_i})e^{\theta' Z_i}$ for $j=N_i+1$. If we use (3), (13) reduces to

$$\bar{S}_W(\theta)=\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{N_i} W\{\tilde{X}_{ij}(\theta)|\theta\}\left[Z_i-\frac{\sum_{k=1}^{n}\sum_{r=1}^{N_k+1} Z_k I\{\tilde{X}_{kr}(\theta)\geqslant \tilde{X}_{ij}(\theta)\}}{\sum_{k=1}^{n}\sum_{r=1}^{N_k+1} I\{\tilde{X}_{kr}(\theta)\geqslant \tilde{X}_{ij}(\theta)\}}\right]. \tag{14}$$

Since (14) is a rank-based estimating function, $\hat{\theta}$ may be defined as a zero-crossing of $\bar{S}_W(\theta)$ or minimiser of $\|\bar{S}_W(\theta)\|$, where $\|v\|=(v'v)^{\frac{1}{2}}$ for a vector v. However, there may

exist several distinct minimisers because $\bar{S}_W(\theta)$ is not necessarily monotone. The use of $W(u|\theta) \equiv \mathbb{S}^{(0)}(u|\theta)$, referred to here as the Gehan weight, mitigates these difficulties and yields a numerically stable, consistent sequence of estimators $\hat{\theta}_G$. However, it does not necessarily yield an estimator with desirable statistical properties.

Other interesting choices include $W(u|\theta) \equiv 1$ and $W(u|\theta) = \hat{F}_0^{\delta}(u)\{1 - \hat{F}_0(u)\}^{\rho}$, where $\hat{F}_0(.)$ is a suitable estimator of $F_0(.)$. The former produces a log-rank-type estimating function, and the latter corresponds to the $G^{\rho,\delta}$-class of log-rank tests in Fleming & Harrington (1991, § 7.2.1). A constant weight is optimal under the accelerated failure time model $\log T = -\theta' Z + \varepsilon$ when ε has an extreme value distribution (Ritov, 1990). If we identify ε with $\log V$, this model is equivalent to $T = Ve^{-\theta' Z}$, where V has a Weibull distribution. This optimality result extends directly to the accelerated gap times model. To be specific, consider $Q(u)$ in (7), and let $\xi > 0$ be a finite constant. Solving $Q(u) = \xi$ for $\lambda_0(.)$ is equivalent to solving the linear differential equation $\dot{\lambda}_0(u) + (1 - \xi)u^{-1}\lambda_0(u) = 0$. Suppose that $\lambda_0(1) = \lambda^{\xi}\xi$ for any $\lambda > 0$. Then, $\lambda_0(u) = \lambda\xi(\lambda u)^{\xi - 1}$, that is the hazard function of a Weibull distribution. Hence the log-rank estimating funtion, hereafter denoted by $S_{\mathrm{LR}}(\theta)$, is optimal for a large and flexible class of baseline intensity functions.

Despite its simplicity, $S_{\mathrm{LR}}(\theta)$ is not guaranteed to be componentwise monotone in θ and, similarly to more general weight functions, has the potential to be ill-behaved. In § 4·1, it is established that the Gehan estimator $\hat{\theta}_G$ is consistent and asymptotically normal under very weak conditions. These results are then used to propose a novel one-step estimator for θ for the case of a general weight function. The proposed estimator has several useful features that are discussed in more detail in § 4·2.

4. Asymptotics and inference

4·1. *Inference for θ_0: Gehan weight*

If we use the Gehan weight $W(u|\theta) = \mathbb{S}^{(0)}(u|\theta)$, (13) reduces to

$$\bar{S}_G(t, \theta) = \int_0^t \mathbb{S}^{(0)}(u|\theta)\left\{n^{-1}\sum_{i=1}^{n} Z_i N_i(du|\theta)\right\} - \int_0^t \mathbb{S}^{(1)}(u|\theta)\left\{n^{-1}\sum_{i=1}^{n} N_i(du|\theta)\right\}. \quad (15)$$

Letting $t \to \infty$, we obtain

$$\bar{S}_G(\theta) = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{N_i}\sum_{k=1}^{n}\sum_{r=1}^{N_k+1}(Z_i - Z_k)I\{\tilde{X}_{kr}(\theta) \geqslant \tilde{X}_{ij}(\theta)\},$$

which is the gradient of the convex objective function

$$L_G(\theta) = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{N_i}\sum_{k=1}^{n}\sum_{r=1}^{N_k+1}\{\log \tilde{X}_{kr}(\theta) - \log \tilde{X}_{ij}(\theta)\}^{+} \quad (16)$$

for $a^{+} = \max\{a, 0\}$; see for example Fygenson & Ritov (1994) or Lin et al. (1998). The minimisers of $L_G(\theta)$ and $\|\bar{S}_G(\theta)\|$ are asymptotically equivalent (Fygenson & Ritov, 1994). Hence we let $\hat{\theta}_G = \arg\min L_G(\theta)$, where the right-hand side is taken to mean any value of θ minimising $L_G(\theta)$. Similarly to Lin et al. (1998), $\hat{\theta}_G$ can be efficiently computed using linear programming methods. Theorem 1 summarises the large sample behaviour of $\hat{\theta}_G$ under Conditions A1–A7 of the Appendix, providing the necessary results for inference.

THEOREM 1. *The estimator* $\hat{\theta}_G = \arg\min L_G(\theta)$ *exists with probability tending to one and* $n^{\frac{1}{2}}(\hat{\theta}_G - \theta_0) \to N(0, \Gamma_G)$ *in distribution, where* $\Gamma_G = D_G^{-1}\Sigma_G D_G^{-1}$,

$$\Sigma_G = \int_0^\infty [\{s^{(0)}(u|\theta_0)\}^2 s^{(2)}(u|\theta_0) - s^{(0)}(u|\theta_0)\{s^{(1)}(u|\theta_0)\}^{\otimes 2}]\,\lambda_0(u)du, \tag{17}$$

$$D_G = -\int_0^\infty [s^{(0)}(u|\theta_0)s^{(2)}(u|\theta_0) - \{s^{(1)}(u|\theta_0)\}^{\otimes 2}]Q(u)\lambda_0(u)du. \tag{18}$$

Notably, Theorem 1 does not impose a bounded gap-time interval assumption, a useful consequence of the convexity associated with the Gehan weight function. A method for consistently estimating Γ_G is proved in § 4·3.

4·2. *Inference for* θ_0: *General weight*

The monotonicity properties of $\bar{S}_G(\theta)$ are valuable both computationally and in proving large sample results. Typically, the benefits derived from monotonicity are lost when a different weight function, such as $W(u|\theta) = 1$, is chosen, the potential gain being in terms of efficiency. These benefits can be partially recovered using a one-step estimator derived from $\hat{\theta}_G$. For example, the asymptotic results of Tsiatis (1990), Ritov (1990), Ying (1993) and Lin et al. (1998) for general weight functions require that θ_0 lie in some known compact set. This assumption, routinely ignored in practical work, is used to ensure that a consistent estimator sequence may be chosen. It need not be imposed for the one-step estimator of Theorem 2 because $\hat{\theta}_G$ is a $n^{\frac{1}{2}}$-consistent estimator in the absence of this assumption. This stems from the convexity of (16), which also provides computational benefits by avoiding the ill-behaved problem of minimising $\|\bar{S}_W(\theta)\|$ directly. Despite these improvements, the large sample theory for a general weight function still requires regularity conditions more restrictive than those of Theorem 1. The following one-step estimation result holds under Conditions A1–A10 of the Appendix.

THEOREM 2. *Let* $\hat{D}_W$ *be any consistent estimator of* D_W *defined in* (20) *below. Then* $\hat{\theta}_W = \hat{\theta}_G - \hat{D}_W^{-1}\bar{S}_W(\hat{\theta}_G)$ *is* $n^{\frac{1}{2}}$*-consistent and* $n^{\frac{1}{2}}(\hat{\theta}_W - \theta_0) \to N(0, \Gamma_W)$ *in distribution, where* $\Gamma_W = D_W^{-1}\Sigma_W D_W^{-1}$,

$$\Sigma_W = \int_0^{\bar{\tau}} w^2(u|\theta_0)v(u|\theta_0)s^{(0)}(u|\theta_0)\lambda_0(u)du, \tag{19}$$

$$D_W = -\int_0^{\bar{\tau}} w(u|\theta_0)v(u|\theta_0)s^{(0)}(u|\theta_0)Q(u)\lambda_0(u)du, \tag{20}$$

$v(u|\theta_0) = s^{(2)}(u|\theta_0)/s^{(0)}(u|\theta_0) - \{s^{(1)}(u|\theta_0)/s^{(0)}(u|\theta_0)\}^{\otimes 2}$, *and* $\bar{\tau}$ *and* $w(u|\theta_0)$ *are respectively defined in Conditions* A8 *and* A9 *in the Appendix.*

If $w(u|\theta_0) \equiv Q(u)$ and $\bar{\tau} \to \infty$ in Theorem 2, then both Σ_W and $-D_W$ equal (8), giving $\Gamma_Q = \Sigma_Q^{-1}$ and hence that $\hat{\theta}_Q$ is a semiparametric efficient estimator, by Proposition 1.

A consistent estimator of D_W is required in order to compute $\hat{\theta}_W$. However, because an estimator of D_W is also needed for estimating Γ_W, this is not a serious computational disadvantage. Computation of $\hat{\theta}_W$ and its variance estimate are dealt with in § 4·3.

Remark. Jin et al. (2003) propose an iterative procedure for obtaining a consistent sequence of regression parameter estimators under the accelerated failure time model. Their approach relies on the successive optimisation of a weighted sequence of convex objective functions similar to (16). The proposed one-step estimator can achieve the same result with much less computation.

4·3. *Estimating Γ_G and Γ_W*

The variances in Theorems 1 and 2 depend on (17) and (19), which are easily estimated, and the matrices D_G and D_W respectively given in (18) and (20). The dependence of D_G and D_W on $Q(.)$, and hence on $\lambda_0(.)$ and $\dot{\lambda}_0(.)$, makes estimation of Γ_G and Γ_W challenging in practice. Huang (2002a) proposed an interesting method of variance estimation in a related rank-regression problem. The following algorithm adapts his method for estimating Γ_G.

ALGORITHM 1.
Step 1. *Compute* $\hat{\theta}_G$ *by minimising* (16) *and estimate* Σ_G *by*

$$\hat{\Sigma}_G = \int_0^\infty [\mathbb{S}^{(0)}(u|\hat{\theta}_G)\mathbb{S}^{(2)}(u|\hat{\theta}_G) - \{\mathbb{S}^{(1)}(u|\hat{\theta}_G)\}^{\otimes 2}]\bar{N}^{(0)}(du|\hat{\theta}_G),$$

where $\bar{N}^{(0)}(.|\theta)$ *is defined in equation* (A2).

Step 2. *For* $k = 1, \ldots, p$, *set* $\tilde{\theta}_{Gk} = \arg\min_\theta \|\bar{S}_G(\theta) - c_k\|$, *where* c_k *is the kth column of* $(n^{-1}\hat{\Sigma}_G)^{\frac{1}{2}}$

Step 3. *Compute* $\hat{\Gamma}_G = (\hat{\Gamma}_G^{\frac{1}{2}})^2$, *where* $\hat{\Gamma}_G^{\frac{1}{2}} = n^{\frac{1}{2}}(\tilde{\theta}_{G1} - \hat{\theta}_G, \ldots, \tilde{\theta}_{Gp} - \hat{\theta}_G)$.

Huang (2002a) refers to this method of variance estimation as 'inverse numerical differentiation'. This method does not estimate D_G directly; rather, it uses the asymptotic linearity of $\bar{S}_G(\theta)$ and its inverse near θ_0 to estimate $\hat{\Gamma}_G$ directly. Under Conditions A1–A5, the consistency of $\hat{\Sigma}_G$ is a consequence of Corollary A1 in the Appendix. That $\hat{\Gamma}_G$ consistently estimates Γ_G is a consequence of Theorem 1 and the arguments used to prove Theorem 4 of Huang (2002a). In § 5, $\hat{\Gamma}_G$ is observed to estimate Γ_G accurately for modest sample sizes.

Algorithm 1 requires $p+1$ minimisations of a perturbed nonsmooth objective function. In the case of the Gehan weight, these minimisation problems are well behaved and can be implemented very efficiently. Algorithm 1 can also be adapted to the case of a general weight function. However, most other choices of $W(.|\theta)$ yield non-monotone estimating functions and the required optimisations are more problematic. For example, in addition to the lack of efficient general algorithms, the minimisers may exhibit unstable behaviour. The use of the one-step estimator $\hat{\theta}_W$ avoids these problems. However, because $\hat{\theta}_W$ requires an approximation for D_W and Algorithm 1 requires $\hat{\theta}_W$ while specifically avoiding the need to approximate D_W, the utility of Algorithm 1 for approximating the variance of $\hat{\theta}_W$ is questionable. It is therefore convenient to have a method for approximating D_W directly.

The method proposed below relies on the asymptotic linearity of $\bar{S}_W(\theta)$ but avoids the indicated conflict by making greater use of the properties of $\hat{\theta}_G$. To be specific, for any $0 < M < \infty$ and $\|\theta - \theta_0\| \leqslant Mn^{-\frac{1}{2}}$, the arguments used to prove Theorem 2 imply that

$$\bar{S}_W(\theta) - \bar{S}_W(\theta_0) = D_W(\theta - \theta_0) + o_p(n^{-\frac{1}{2}}). \tag{21}$$

Conditionally on the data, let $\tilde{\theta}_k \sim N(\hat{\theta}_G, n^{-1}\hat{\Gamma}_G)$, where $k = 1, \ldots, B$ and the replications are independent. Then, applying the above identity twice, we informally have

$$\bar{S}_W(\tilde{\theta}_k) - \bar{S}_W(\hat{\theta}_G) = D_W(\tilde{\theta}_k - \hat{\theta}_G) + \text{error}_k \quad (k = 1, \ldots, B).$$

This suggests estimating D_W by the slope in a regression of $\bar{S}_W(\tilde{\theta}_k) - \bar{S}_W(\hat{\theta}_G)$ on $\tilde{\theta}_k - \hat{\theta}_G$ for $k = 1, \ldots, B$. To be more precise, D_W can be estimated with $\tilde{D}_{W,B} = [(\mathfrak{Z}'\mathfrak{Z})^{-1}\mathfrak{Z}'\mathfrak{Y}]_{-1}$, where

$$\mathfrak{Z} = \begin{bmatrix} 1 & (\tilde{\theta}_1 - \hat{\theta}_G)' \\ \vdots & \vdots \\ 1 & (\tilde{\theta}_B - \hat{\theta}_G)' \end{bmatrix}, \quad \mathfrak{Y} = \begin{bmatrix} \{\bar{S}_W(\tilde{\theta}_1) - \bar{S}_W(\hat{\theta}_G)\}' \\ \vdots \\ \{\bar{S}_W(\tilde{\theta}_B) - \bar{S}_W(\hat{\theta}_G)\}' \end{bmatrix}$$

and $[M]_{-1}$ denotes a matrix M with its first row removed.

THEOREM 3. *Let $\hat{D}_W = \lim_{B\to\infty} \tilde{D}_{W,B}$. Then, under Conditions* A1–A10 *of the Appendix, $\hat{D}_W \to D_W$ in probability and $\hat{\Gamma}_W = \hat{D}_W^{-1}\hat{\Sigma}_W\hat{D}_W^{-1}$ is a consistent estimator for*

$$\Gamma_W = D_W^{-1}\Sigma_W D_W^{-1},$$

where

$$\hat{\Sigma}_W = \int_0^{\bar{\tau}} W^2(u|\hat{\theta}_W)\left[\frac{\mathbb{S}^{(2)}(u|\hat{\theta}_W)}{\mathbb{S}^{(0)}(u|\hat{\theta}_W)} - \left\{\frac{\mathbb{S}^{(1)}(u|\hat{\theta}_W)}{\mathbb{S}^{(0)}(u|\hat{\theta}_W)}\right\}^{\otimes 2}\right]\bar{N}^{(0)}(du|\hat{\theta}_W).$$

The proof of this result may be found in the Appendix. Since $\tilde{D}_{W,B}$ is not guaranteed to be symmetric, the symmetric approximation $\hat{D}_{W,B} = \frac{1}{2}(\tilde{D}_{W,B} + \tilde{D}'_{W,B})$ can be employed instead. The main computational algorithm for $\hat{\theta}_W$ and $\hat{\Gamma}_W$ is now summarised.

ALGORITHM 2.
Step 1. *Use Algorithm* 1 *to obtain $\hat{\theta}_G$ and $\hat{\Gamma}_G$.*
Step 2. *Compute $\hat{D}_{W,B}$ for sufficiently large B and subsequently $\hat{\theta}_W = \hat{\theta}_G - \hat{D}_{W,B}^{-1}\bar{S}_W(\hat{\theta}_G)$.*
Step 3. *Compute $\hat{\Gamma}_W$ as in Theorem* 3, *using $\hat{D}_{W,B}$ in place of $\hat{D}_W$.*

The main advantage of this approach to variance approximation stems from the replacement of $p+1$ potentially unstable optimisations with $p+1$ convex optimisations, i.e. for computing $\hat{\theta}_G$ and $\hat{\Gamma}_G$, followed by a simple least squares calculation. The main disadvantage is the additional computation required for calculating $\bar{S}_W(\tilde{\theta}_k)$, for $k = 1, \ldots, B$.

4·4. *Inference for Λ_0*

We consider the general weighted case, that is, where $\Lambda_0(t)$ is estimated using $\hat{\Lambda}_0(.|\hat{\theta}_W)$ and $\hat{\theta}_W$ satisfies Theorem 2. Conditions A1–A10 of the Appendix are therefore assumed. Conditions A9 and A10 can be omitted if $\hat{\theta}_W$ is replaced by $\hat{\theta}_G$.

THEOREM 4. *Let $\hat{\theta}_W$ satisfy Theorem* 2. *Then $n^{\frac{1}{2}}\{\hat{\Lambda}_0(.|\hat{\theta}_W) - \Lambda_0(.)\}$ converges weakly to a Gaussian process on $[0, \bar{\tau}]$ with covariance function*

$$\sigma^2_{\Lambda_0}(s, t) = \int_0^{t\wedge s} \frac{\lambda_0(u)}{s^{(0)}(u|\theta_0)}\,du + A'_Q(t)\Gamma_W A_Q(s), \tag{22}$$

where $A_Q(t)$ is defined in (10). *Moreover, for any* $0 < M < \infty$,

$$\sup_{\substack{t \in [0,\bar{\tau}] \\ n^{1/2}\|\theta-\theta_0\| \leqslant M}} |\hat{\Lambda}_0(t|\theta) - \hat{\Lambda}_0(t|\theta_0) + A_Q'(t)(\theta-\theta_0)| = o_p(n^{-\frac{1}{2}}). \tag{23}$$

If $w(u|\theta_0) = Q(u)$, Theorem 2 shows that Γ_W reduces to Σ_Q^{-1}, the latter defined in (8). Proposition 2 and Theorem 4 therefore imply that efficient estimation of $\Lambda_0(.)$ requires $\hat{\theta}_W$ to be asymptotically equivalent to $\hat{\theta}_Q$.

The variance (22) for $s = t$ can be consistently estimated by

$$\hat{\sigma}^2_{\Lambda_0}(t) = \int_0^t \{\mathbb{S}^{(0)}(u|\hat{\theta}_W)\}^{-2} \bar{N}^{(0)}(du|\hat{\theta}_W) + \hat{A}'(t)\hat{\Gamma}_W \hat{A}(t),$$

where $\hat{\Gamma}_W$ is estimated as in Theorem 3 and $\hat{A}(t)$ is any consistent estimator of $A_Q(t)$. The linear expansion (23) shows that $A_Q(.)$ can be approximated in a manner similar to Theorem 3. Confidence intervals and bands for $\Lambda_0(t)$ can then be computed in standard fashion. For example, a $100(1-\alpha)\%$ confidence interval for $\Lambda_0(t)$ is (Andersen et al., 1993, § IV.1.3)

$$\hat{\Lambda}_0(t|\hat{\theta}_W) \exp[\mp z_{\alpha/2} \hat{\sigma}_{\Lambda_0}(t)/\{n^{\frac{1}{2}} \hat{\Lambda}_0(t|\hat{\theta}_W)\}].$$

5. Simulation results

Two simulations were done to evaluate the performance of $\hat{\theta}_G$ and also $\hat{\theta}_W$ for the log-rank weight $W(.|\theta) = 1$, hereafter denoted by $\hat{\theta}_{\mathrm{LR}}$. In the first simulation, $T \sim Ve^{-0{\cdot}5Z}$, where $V \sim \mathrm{Ex}(1)$, $Z \sim \mathrm{Ber}(0{\cdot}5)$, and $\tau \sim \mathrm{Un}(0, 3{\cdot}5)$. This setting is equivalent to the 'Poisson process' simulation of Lin et al. (1998, Table 1). In the second simulation, $V \sim \mathrm{Ga}(0{\cdot}75, 0{\cdot}75)$ and $Z \sim N(0, 1)$; T is then computed as above, with $\tau \sim \mathrm{Un}(0, 3{\cdot}5)$. Sample sizes of $n = 50$ and $n = 100$ are considered, and 1000 replications are used. In both cases, $\theta_0 = 0{\cdot}5$; $\{\Lambda_0(1), \Lambda_0(2)\}$ equal $(1, 2)$ and $(1{\cdot}73, 3{\cdot}19)$, respectively.

The simulations were done in MATLAB 6; the results are summarised in Table 1. The results for the log-rank and Gehan weights for a given combination of baseline distribution and sample size, i.e. columns (2, 6), (3, 7), (4, 8) and (5, 9), are based on the same 1000 datasets. The estimators $\hat{\theta}_G$ and $\hat{\Gamma}_G$ are obtained using Algorithm 1, and $\hat{\theta}_{\mathrm{LR}}$ and $\hat{\Gamma}_{\mathrm{LR}}$ are obtained using Algorithm 2. Empirical biases and standard errors are reported in the table. For comparison, a version of Algorithm 1 modified for the log-rank weight and with $\hat{\theta}_{\mathrm{LR}}$ replacing $\hat{\theta}_G$ is also used to approximate Γ_W. The empirical mean of the Huang-type variance estimators for $\hat{\theta}_G$ and $\hat{\theta}_{\mathrm{LR}}$ are given under '$E\{\mathrm{ESE}(\hat{\theta})\}$ (Huang)'. For the log-rank weight, '$E\{\mathrm{ESE}(\hat{\theta})\}$ (Thm 3)' refers to approximations based on Theorem 3; these were not computed for the Gehan weight. In estimating these standard errors, we respectively use $B = 100$ and $B = 250$ simulated values per dataset for $\hat{\theta}_{\mathrm{LR}}$ and $\{\hat{\Lambda}(.|\hat{\theta}_G), \hat{\Lambda}(.|\hat{\theta}_{\mathrm{LR}})\}$.

The proposed estimation methods work quite well for practical sample sizes. All estimators are nearly unbiased and confidence interval coverage agrees nicely with the nominal level of 95%. In the case of the log-rank weight, there is no evidence that using all the data creates bias, which suggests that one may let $\bar{\tau} \to \infty$ in Theorem 2. The standard error of $\hat{\theta}_{\mathrm{LR}}$ decreases when moving from Exponential to Gamma gap times for

Table 1. *Summary of simulation studies on the estimation of θ_0 and $\Lambda_0(t)$*

	Log-rank				Gehan			
	Exponential		Gamma		Exponential		Gamma	
	$n=50$	$n=100$	$n=50$	$n=100$	$n=50$	$n=100$	$n=50$	$n=100$
$\lvert\text{Bias}(\hat{\theta})\rvert$	0·004	0·001	0·000	0·002	0·007	0·006	0·003	0·004
$\text{SE}(\hat{\theta})$	0·209	0·144	0·095	0·065	0·237	0·160	0·113	0·076
$E\{\text{ESE}(\hat{\theta})\}$ (Huang)	0·192	0·135	0·092	0·063	0·227	0·158	0·110	0·076
$E\{\text{ESE}(\hat{\theta})\}$ (Thm 3)	0·209	0·141	0·097	0·065	—	—	—	—
Coverage (95%)	0·944	0·943	0·957	0·955	0·945	0·949	0·944	0·950
$\lvert\text{Bias}\{\hat{\Lambda}(1)\}\rvert$	0·000	0·006	0·001	0·004	0·006	0·000	0·002	0·001
$\text{SE}\{\hat{\Lambda}_0(1)\}$	0·172	0·118	0·189	0·130	0·183	0·127	0·190	0·134
$E[\text{ESE}\{\hat{\Lambda}_0(1)\}]$	0·168	0·117	0·181	0·126	0·177	0·126	0·184	0·129
Coverage (95%)	0·938	0·958	0·953	0·947	0·933	0·957	0·950	0·944
$\lvert\text{Bias}\{\hat{\Lambda}(2)\}\rvert$	0·008	0·000	0·003	0·003	0·003	0·003	0·034	0·015
$\text{SE}\{\hat{\Lambda}_0(2)\}$	0·388	0·271	0·532	0·390	0·394	0·276	0·514	0·388
$E[\text{ESE}\{\hat{\Lambda}_0(2)\}]$	0·375	0·263	0·523	0·368	0·381	0·274	0·520	0·372
Coverage (95%)	0·950	0·948	0·944	0·947	0·952	0·959	0·948	0·942

SE, standard error; ESE, estimated standard error.

a given sample size; this reflects an increase in the expected number of events per subject from 2·3 to 3·7. In contrast, the variances of the cumulative hazard estimators increase, reflecting a reduction in the expected number at risk in the case of Gamma-distributed gap times at times $t=1, 2$. Interestingly, the standard errors of the variance estimators obtained using Algorithm 2 never exceeded those of Huang's estimator, despite the need to use simulation in computing the former; the results are not shown.

6. BLADDER TUMOUR DATA ANALYSIS

In this section we illustrate the proposed methods using the bladder tumour cancer data of Byar (1980). The data come from a randomised clinical trial conducted by the Veterans Administration Co-operative Urological Group from 1971–1976 and consist of 118 patients with bladder tumours. These tumours were removed, and patients were then randomised to one of three treatments, placebo, pyridoxine or thiotepa. The data on thiotepa, for 38 patients and 45 total recurrences, versus placebo, for 48 patients and 87 total recurrences, are analysed in Lin et al. (1998), where they evaluate the effect of the baseline variables treatment, scored as 1 for placebo and 0 for thiotepa, number of initial tumours removed, scored as an integer with range 1–8, and diameter of largest initial tumour, ranging from 1 to 8 centimetres, on the average number of recurrences. Kalbfleisch & Prentice (2002, §§ 9.4.3, 9.5.2) also analyse these data using several intensity and marginal models. Their analyses suggest that the recurrence rate depends on the sojourn time and also that the risk of censoring is elevated for those patients experiencing a tumour recurrence within the past month. Consequently, the data were reanalysed under the accelerated gap times model using only baseline covariates. Table 2 contains the results. The estimates are obtained using Algorithms 1 and 2, with $B=1500$ simulated values being used in Step 2 of Algorithm 2 for estimating $\hat{D}_{\text{LR}}$.

If we recall that the model assumes that $T_j = V_j e^{-\theta_0' Z}$ for $j \geqslant 1$, the results suggest that the times between tumour recurrences are expanded on treatment and contract as the number of initial tumours increases. The treatment effect and initial number of tumours are both

Table 2. *Analysis of the bladder tumour data*

Covariate	Log-rank weight $\hat{\theta}$	Log-rank weight Est. SE	Gehan weight $\hat{\theta}$	Gehan weight Est. SE
Treatment	0·681*	0·278	0·433	0·257
Initial number	0·264*	0·067	0·207*	0·064
Initial size	−0·031	0·096	−0·008	0·090

Est. SE; estimated standard error.
*Significant at the 5% level.

statistically significant at the 5% level. These results are similar to those obtained for the model of Lin et al. (1998, Table 2). The estimated standard errors under both models are also similar, though appreciably smaller for the treatment effect under the accelerated gap times model. The smaller standard errors obtained using the Gehan weight function may suggest that a decreasing rather than constant weight function is a more appropriate choice for these data.

The accelerated gap times model allows estimation of the recurrence time distribution (2) and quantities derived therefrom. Estimates of (2) using the log-rank weight by treatment arm for the 'average' patient, i.e. with 2·5 initial tumours, the largest being 1·94 centimetres, are given in Fig. 1; also provided are the estimated median times to tumour recurrence. The observed ordering may be partly an artifact because the estimated distribution for the placebo group is obtained from that for the treatment group by a monotone time transformation. The estimated median time to tumour recurrence for the treated group is nearly double that for the placebo group, reflecting the effectiveness of thiotepa at

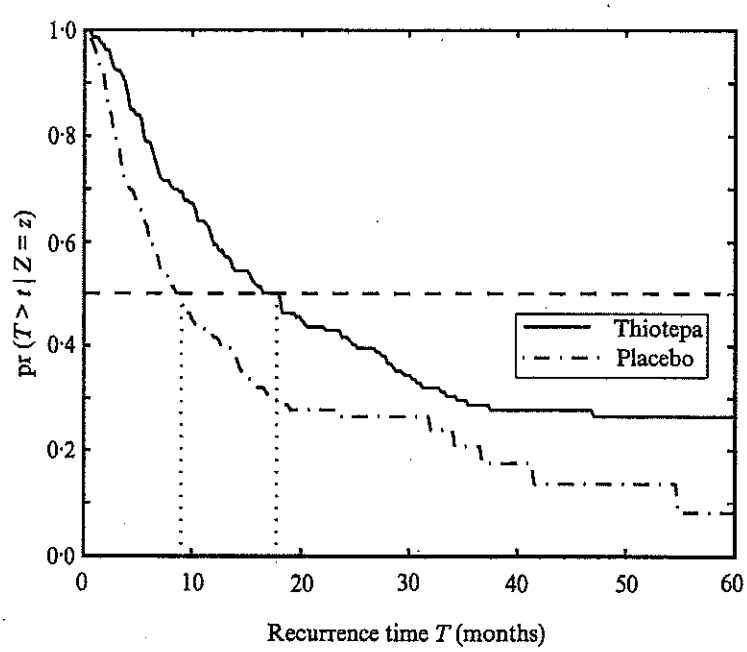

Fig. 1: Bladder tumour data. Estimated recurrence time distributions by treatment. Dotted vertical lines indicate median (placebo) = 9·0 and median (thiotepa) = 17·8.

reducing tumour recurrence. This is also reflected in the regression coefficient for the treatment effect, which suggests that recurrence times for patients on thiotepa should be $\exp(0{\cdot}681) \simeq 2$ times as long as for patients on placebo.

For a hypothetical patient with fixed covariates $\tilde{z}$, we can easily obtain a confidence interval for the median time to recurrence. Let $\tilde{\theta}_W$ and $\tilde{\Lambda}_0(t|\tilde{\theta}_W)$ denote the regression parameter and associated cumulative baseline intensity estimates based on the data $\{N_i^\dagger(u), Y_i^\dagger(u), \tilde{Z}_i, u \geqslant 0\}$, for $i = 1, \ldots, n$, where $\tilde{Z}_i = Z_i - \tilde{z}$. If we use $\tilde{Z}$, $\Lambda_0(t)$ corresponds to a patient having covariates $\tilde{z}$. A family of 95% confidence intervals for the median recurrence time of a patient with covariates $\tilde{z}$ is then (Andersen et al., 1993, § IV.3.4)

$$\left\{\xi \geqslant 0 : \frac{|g\{\tilde{\Lambda}_0(\xi|\tilde{\theta}_W)\} - g(\log 2)|}{|\dot{g}\{\tilde{\Lambda}_0(\xi|\tilde{\theta}_W)\}|\tilde{\sigma}_n(\xi)} \leqslant 1{\cdot}96\right\},$$

where $g(.)$ is any suitable monotone transformation and $\tilde{\sigma}_n(.)$ is the estimated standard deviation of $\tilde{\Lambda}_0(.|\tilde{\theta}_W)$. For example, with $g(x) = \log x$, 95% confidence intervals for the median time to recurrence are (10·5, 27·0) and (6·5, 12·5) months for an average patient treated with thiotepa or placebo, respectively.

7. REMARKS

Few results exist on efficient estimation for intensity models except for the class of multiplicative intensity models. One exception is Chang & Hsiung (1994), who study estimation of the regression parameter in a class of model including the modulated renewal process. Estimation, asymptotics and efficiency theory for (θ_0, Λ_0) under the modulated renewal process model can be handled using the tools of this paper, generalising the results of Oakes & Cui (1994) to data involving multiple censored realisations of renewal processes.

The use of a one-step estimator for general weight functions is inspired by Bickel et al. (1993, Ex. 7.7.2), who consider the accelerated failure time model. No guidance is provided there on the required step of estimating the proper analogue of D_W. For estimating variances, Jones (1997) suggests approximating this quantity using linear regression, creating replicates by constructing a grid of fixed bandwidth about $\hat{\theta}$. An important disadvantage of this approach is the difficulty in selecting a proper grid bandwidth. The proposed method relies mainly on the fact that (21) remains valid for $\theta = \hat{\theta}_G$, addressing the grid bandwidth problem by making use of the asymptotic properties of $\hat{\theta}_G$. The resulting methodology is appealingly simple and performs well in practice. Generating $\tilde{\theta}_k$ from a $N(\hat{\theta}_G, n^{-1}\hat{\Gamma}_G)$ distribution, while not required, is perfectly natural in view of Theorem 1.

The accelerated gap times model is most appropriate for recurrent event data of the form $U_0 < U_1 < \ldots$ with U_0 corresponding to an event occurrence. If U_0 corresponds to some other entry time, then the distribution of $T_1 = U_1$ will typically differ from T_k for $k \geqslant 2$. In fact, the resulting delayed renewal process (Daley & Vere-Jones, 1988, p. 71) can be handled within the context of the accelerated gap times model by allowing the baseline intensity associated with the first event time to be different from that of the remaining gap times. The methods and results of this paper are easily extended to this setting.

The accelerated gap times model has two important limitations. First, the dependence of each gap time on the previous event and covariate history must be correctly specified. In this respect the renewal assumption imposed by the use of baseline covariates is

rather strong. The use of time-dependent covariates permits significant relaxation of the renewal assumption by allowing specific forms of gap-time dependence, such as Markov dependence. There are several ways of extending the accelerated gap times model to the case of time-dependent covariates, the most natural perhaps being obtained by the extension of the accelerated failure time model considered in Robins & Tsiatis (1992). Aalen et al. (2004) make a strong case for using time-dependent covariates in the context of intensity modelling. Of course, capturing the precise nature of the dependence using time-dependent covariates may be difficult, and in this regard marginal models can be useful in cases where population quantities that are invariant to this dependence, such as means, are of primary interest. A second limitation of the accelerated gap times model is that it assumes all events to be of the same type. Neither Lin et al. (1998) nor Huang (2002b) impose this assumption, with the latter being a more useful marginal model for analysing data involving multiple event types. A limitation of both marginal models is the inability to study factors affecting transitions between states. Such questions are better suited to an intensity model, which requires one to specify how these transitions depend on the past event and state transition histories.

Several other possibilities exist for extension of the accelerated gap times model. For example, Prentice et al. (1981) allow for event-dependent stratification of the baseline intensity $\lambda_0(.)$ under the modulated renewal process model. It is not clear how to extend the efficiency theory of this paper to a similar extension of the accelerated gap times model. However, it is possible to construct an unbiased estimating function for θ by repeating the calculations leading to Proposition 1. The properties of such an estimator, as well as methods for variance estimation, merit further investigation. In the absence of covariates, Aalen & Husebye (1991) propose to model the gap times using a linear mixed model with two variance components. This suggests considering a linear mixed effects model for $\log T_{ij}$, for $j \geqslant 1$ and $i = 1, \ldots, n$, with covariate- and subject-specific random effects, allowing for clustered recurrent event processes, such as patients treated at the same hospital or by the same physician, in addition to within-subject dependence. Conditionally on the random effects, the intensity formulation of such a model coincides with the accelerated gap times model. In principle, inference for the fixed effects and variance components can then be derived from the implied marginal likelihood. A more common method for handling dependence is to incorporate a frailty into the intensity model. Peña et al. (2001) implemented the shared frailty model for recurrent event data in the absence of covariate information, deriving semiparametric estimators for the gap-time distribution assuming exchangeable rather than independent gap times. One may instead consider using an appropriate modification of the correlated gamma frailty model considered in Parner (1998) in order to allow additionally for constant levels of correlation between cluster members. Compared to the mixed effects formulation mentioned above, these frailty models provide greater mathematical tractability in exchange for less flexibility in modelling the dependence structure.

Acknowledgement

This work was supported by the U.S. National Institutes of Health. The author thanks Edsel Peña for commenting on an earlier draft, Jon Wellner for suggestions on the proof of Theorem A2, and the editor, an associate editor and two referees for their helpful comments.

APPENDIX

Technical details

Regularity conditions. The regularity conditions listed below are used in proving various results in this paper.

Condition A1. We require that $F_0(.) = \int_0^{\cdot} f_0(s)ds$, where $f_0(.)$ has continuous first and bounded second derivatives.

Condition A2. We require that $\{N_i^{\dagger}(u), Y_i^{\dagger}(u), Z_i, u \geqslant 0\}$, for $i = 1, \ldots, n$, be independent and identically distributed.

Condition A3. We require that $\tau_1 | Z_1$ have a possibly degenerate distribution $G(.|Z_1)$.

Condition A4. Censoring is noninformative.

Condition A5. We require that Z_1 be almost surely bounded and $E[\{N_1^v(\tau_1)\}^{6+\varepsilon}] < \infty$ for some $\varepsilon > 0$.

Condition A6. We require that $s^{(j)}(u|\theta)$, for $j = 0, 1, 2$, be continuous for $(u, \theta) \in \mathbb{R}^+ \times \mathcal{N}(\theta_0)$, where $\mathcal{N}(\theta_0)$ is some neighbourhood of θ_0.

Condition A7. We require that D_G in (18) be nonsingular.

Condition A8. We require that $\bar{\tau}$ satisfy $\inf\{t \in [0, \bar{\tau}] : s^{(0)}(t|\theta_0)\} > 0$.

Condition A9. For $(t, \theta) \in [0, \bar{\tau}] \times \mathcal{N}(\theta_0)$, $W(t|\theta)$ is of bounded variation and there exists a continuous, bounded deterministic function $w(t|\theta)$ such that $|W(t|\theta) - w(t|\theta)|$ converges to zero uniformly in probability.

Condition A10. We require that D_W in (20) be nonsingular.

Proof of Proposition 1. For all $\eta \in \mathbb{H}$, we must determine $\varphi_k(.) \in \mathbb{H}$ $(k = 1, \ldots, p)$ such that $E[\{S_{Qk}(\theta_0) - B_{\theta_0,\Lambda_0}(\varphi_k)\} B_{\theta_0,\Lambda_0}(\eta)] = 0$. It suffices to consider $k = 1$ and $\eta(.)$ not identically zero. Both $S_{Q1}(\theta_0)$ and $B_{\theta_0,\Lambda_0}(\eta)$ are stochastic integrals with respect to $M_i^{\dagger}(.)$, for $i = 1, \ldots, n$, of Lemma 1. Since $Q(.)$ and $\eta(.)$ are deterministic, a martingale covariance calculation and subsequent use of Lemma 2 give

$$E[B_{\theta_0,\Lambda_0}(\eta)\{S_{Q1}(\theta_0) - B_{\theta_0,\Lambda_0}(\varphi_1)\}] \equiv nE\left[\int_0^{\infty} \{Z_{11}Q(r) - \varphi_1(r)\}\eta(r)\lambda_0(r)Y_1(r|\theta_0)dr\right].$$

Let $\mathscr{E}_1(u) = E\{Z_{11}Y_1(r|\theta_0)\}/E\{Y_1(r|\theta_0)\}$. Then $E[B_{\theta_0,\Lambda_0}(\eta)\{S_{Q1}(\theta_0) - B_{\theta_0,\Lambda_0}(\varphi_1)\}] = 0$ if and only if $\varphi_1(r) = Q(r)\mathscr{E}_1(u)$. Since Z_1 is uniformly bounded by Condition A5 and $\mathscr{E}_1(r) = 0$ if $E\{Y_1(r|\theta_0)\} = 0$, it follows that $\varphi_1(.) \in \mathbb{H}$. The efficient score is obtained by subtracting $[B_{\theta_0,\Lambda_0}(\varphi_1), \ldots, B_{\theta_0,\Lambda_0}(\varphi_p)]'$ from $S_Q(\theta_0)$; (7) and (8) then follow from Lemma 3. The information bound is now a consequence of Theorem 3.4.1 of Bickel et al. (1993). □

Proof of Proposition 2. The efficient score for $\Lambda_0(.)$ equals

$$B_{\theta_0,\Lambda_0}(\eta_{\text{eff},t}) - E\{B_{\theta_0,\Lambda_0}(\eta_{\text{eff},t})S_Q'(\theta_0)\}(n\Sigma_Q)^{-1}S_{\text{eff}}(\theta_0), \tag{A1}$$

where $B_{\theta_0,\Lambda_0}(\eta_{\text{eff},t}) = \sum_{i=1}^n \int_0^{\infty} [I\{R_i(u)e^{\theta_0'Z_i} \leqslant t\}/s^{(0)}\{R_i(u)e^{\theta_0'Z_i}|\theta_0\}]M_i^{\dagger}(du)$, $S_Q(\theta_0)$ is given in (5), $S_{\text{eff}}(\theta_0)$ is given in Proposition 1, and Σ_Q is defined in (8) (van der Vaart, 1998, § 25.5). A martingale covariance calculation and use of Lemma 2 yields $E\{B_{\theta_0,\Lambda_0}(\eta_{\text{eff},t})S_Q(\theta_0)\} = nA_Q(t)$, where $A_Q(t)$ is defined in (10). Hence, (A1) reduces to

$$B_{\theta_0,\Lambda_0}^{(\text{eff})}(t) = \sum_{i=1}^n \int_0^{\infty} \left[\frac{I\{u \leqslant t\}}{s^{(0)}(u|\theta_0)} - A_Q'(t)\Sigma_Q^{-1}Q(u)\{Z_i - \mathscr{E}(u)\}\right] M_i(du|\theta_0).$$

By Lemma 3 and the fact that the summands are independent and identically distributed, we have

$$\operatorname{var}\{n^{-1/2}B^{(\mathrm{eff})}_{\theta_0,\Lambda_0}(t)\} = E\left(\int_0^\infty \left[\frac{I\{u \leqslant t\}}{s^{(0)}(u|\theta_0)} - A'_Q(t)\Sigma_Q^{-1}Q(u)\{Z_1 - \mathscr{E}(u)\}\right]^2 Y_1(u|\theta_0)\lambda_0(u)du\right).$$

Expanding the square and simplifying yields (11). □

We now provide proofs and auxiliary results for § 4. We begin with several fundamental results. All probabilities and expectations are with respect to $\mathbb{P}_0$, the probability measure associated with θ_0 and $\Lambda_0(.)$. Throughout, $\mathbb{R}^+ = [0, \infty)$, and Θ is an open convex subset of $\mathbb{R}^p$. The proofs for Theorems A1 and A2 are available in a technical report and use the empirical process theory developed in van der Vaart & Wellner (1996). The proofs for results in § 4 are collected at the end.

THEOREM A1. *Let $\theta \in \Theta$ be fixed and define, for $t \in \mathbb{R}^+$,*

$$\bar{N}^{(j)}(t|\theta) = \frac{1}{n}\sum_{i=1}^n Z_i^{\otimes j} N_i(t|\theta) \quad (j = 0, 1, 2). \tag{A2}$$

Then, considered as processes in $t \in \mathbb{R}^+$, (12) and (A2) are $\mathbb{P}_0$-Donsker.

THEOREM A2. *As processes in $(t, \theta) \in \mathbb{R}^+ \times \Theta$, (12) and (A2) are $\mathbb{P}_0$-Donsker.*

The following an immediate consequence of Theorems A1 and A2.

COROLLARY A1. *Under Conditions A1–A5 and for $j = 0, 1, 2$,*

$$\sup_{(t,\theta)\in\mathbb{R}^+\times\Theta} \|\bar{N}^{(j)}(t|\theta) - E\{Z_1^{\otimes j}N_1(t|\theta)\}\| \to 0, \qquad \sup_{(t,\theta)\in\mathbb{R}^+\times\Theta} \|\bar{Y}^{(j)}(t|\theta) - E\{Z_1^{\otimes j}Y_1(t|\theta)\}\| \to 0,$$

both in probability. These results continue to hold if the supremum is taken only over $t \in \mathbb{R}^+$, with $\theta \in \Theta$ fixed.

Lemma A1 below is useful for computing the expectations appearing in Corollary A1 and is an easy consequence of (4), (3) and Lemma 3. The proof is therefore omitted.

LEMMA A1. *Let $\theta \in \mathbb{R}^p$ be arbitrary. Then $N_i(t|\theta) = N_i(te^{(\theta_0-\theta)'Z_i}|\theta_0)$ and $Y_i(t|\theta) = Y_i(te^{(\theta_0-\theta)'Z_i}|\theta_0)$, for $i = 1, \ldots, n$. Moreover, for $k = 0, 1, 2$,*

$$E\{Z_1^{\otimes k}N_1(t|\theta)\} = E\left\{Z_1^{\otimes k}\int_0^{te^{(\theta_0-\theta)'Z_1}} Y_1(u|\theta_0)\lambda_0(u)du\right\}.$$

Proof of Theorem 1. Under Conditions A1–A6, Corollary A1, (15) and Lemma A1 imply that $\bar{S}_G(\theta)$ converges uniformly to

$$\mathscr{S}_G(\theta) = \int_0^\infty (s^{(0)}(u|\theta)E[Z_1 Y_1(u|\theta)\lambda_0\{uh_1(\theta)\}h_1(\theta)] - s^{(1)}(u|\theta)E[Y_1(u|\theta)\lambda_0\{uh_1(\theta)\}h_1(\theta)])du,$$

where $h_1(\theta) = \exp\{(\theta_0 - \theta)'Z_1\}$. Since $h_1(\theta_0) = 1$, $\mathscr{S}_G(\theta_0) = 0$; Conditions A6 and A7 then ensure that θ_0 is unique. Corollary A1 and Theorem A1 imply that $\bar{S}_G(\theta_0) = \bar{\zeta}_n + o_p(n^{-1/2})$ for $\zeta_i = \int_0^\infty \{s^{(0)}(u|\theta_0)Z_i - s^{(1)}(u|\theta_0)\}M_i(du|\theta_0)$. Using Lemma 3, we can show that $\operatorname{var}(\zeta_i) = \Sigma_G$. By Conditions A1 and A6, $\|\Sigma_G\| < \infty$, and hence $\bar{\zeta}_n$ satisfies an ordinary central limit theorem.

Define $D_G(\theta) = (d/d\theta)\mathscr{S}_G(\theta)$; then Conditions A1–A6 imply that $D_G(\theta)$ exists and is continuous at $\theta = \theta_0$. By (A7), $D_G = D_G(\theta_0)$ in (18) is nonsingular. Define $C_M = \{\alpha \in \mathbb{R}^p : \|\alpha\| \leqslant M\}$, where $0 < M < \infty$ is arbitrary. Since $\bar{S}_G(\theta)$ and $\mathscr{S}_G(\theta)$ are componentwise monotone in θ, the result follows from Theorem 24.2 of Ritov (1991) if, for any positive sequence $\delta_n \to 0$,

$$\sup_{\|\theta-\theta_0\|\leqslant\delta_n} n^{1/2}\{\bar{S}_G(\theta) - \bar{S}_G(\theta_0) - \mathscr{S}_G(\theta)\} = o_p(1 + n^{1/2}\|\theta - \theta_0\|). \tag{A3}$$

To show (A3), we first write $\bar{S}_G(\theta) = \bar{S}_G(\theta_0) + \bar{\Delta}(\theta, \theta_0)$, where $\bar{\Delta}(\theta, \theta_0) = \bar{S}_G(\theta) - \bar{S}_G(\theta_0)$. Using the representations for $N_i(t|\theta)$ and $Y_i(t|\theta)$ in Lemma A1, we can show that

$$\bar{\Delta}(\theta, \theta_0) = \frac{1}{n} \sum_{i=1}^{n} \int_0^\infty \{\mathbb{S}^{(0)}(u|\theta)Z_i - \mathbb{S}^{(1)}(u|\theta)\}Y_i(u|\theta)[\lambda_0\{uh_i(\theta)\}h_i(\theta) - \lambda_0(u)]du.$$

Now, suppose that $\|\theta - \theta_0\| \leqslant \delta_n$, where $\delta_n \to 0$; then Condition A1 implies that

$$\lambda_0\{uh_i(\theta)\}h_i(\theta) = \lambda_0(u) - q_0(u)Z_i'(\theta - \theta_0) + O(\|\theta - \theta_0\|^2)$$

almost surely, where $q_0(u) = \lambda_0(u) + u\dot{\lambda}_0(u)$ and the constant in the error term can be bounded independently of δ_n. Substituting this expansion into $\mathscr{S}_G(\theta)$, we have

$$\mathscr{S}_G(\theta) = -(\theta - \theta_0)' \int_0^\infty [s^{(0)}(u|\theta)s^{(2)}(u|\theta) - \{s^{(1)}(u|\theta)\}^{\otimes 2}]q_0(u)du + o(\|\theta - \theta_0\|). \tag{A4}$$

Substituting the same expansion into $\bar{\Delta}(\theta, \theta_0)$ and then subtracting (A4), we can use Theorem A2, the Delta Method and the fact that $\delta_n \to 0$ to obtain (A3). Observing that (18) is a direct consequence of (A4) and that $q_0(u) = Q(u)\lambda_0(u)$ now completes the proof. □

Proof of Theorem 2. Similarly to the proof of Theorem 1, the stated conditions imply that $\bar{S}_W(\theta) = \bar{S}_W(\bar{\tau}, \theta)$ converges uniformly in probability, for $\theta \in \mathscr{N}(\theta_0)$, to a limit $\mathscr{S}_W(\theta)$ such that $\mathscr{S}_W(\theta_0) = 0$, with Conditions A6 and A10 ensuring that θ_0 is unique. Using Corollary A1, Theorem A1 and Lemma 3, we can write $\bar{S}_W(\theta_0) = \bar{\kappa}_n + o_p(n^{-1/2})$, where

$$\kappa_i = \int_0^{\bar{\tau}} w(u|\theta_0)[Z_i - \{s^{(1)}(u|\theta_0)/s^{(0)}(u|\theta_0)\}]M_i(du|\theta_0)$$

has variance Σ_W satisfying $\|\Sigma_W\| < \infty$. Hence $\bar{\kappa}_n$ satisfies an ordinary central limit theorem.

Define $D_W(\theta) = (d/d\theta)\mathscr{S}_W(\theta)$; Conditions A1–A9 then imply that $D_W(\theta)$ exists and is continuous at $\theta = \theta_0$ and Condition A10 asserts that $D_W = D_W(\theta_0)$ in (20) is nonsingular. Since $\hat{\theta}_G$ is $n^{1/2}$-consistent, Theorem 7.3.2 of Bickel et al. (1993) now implies the result if

$$\sup_{\|\theta - \theta_0\| \leqslant \delta_n} n^{1/2}\{\bar{S}_W(\theta) - \bar{S}_W(\theta_0) - \mathscr{S}_W(\theta)\} = o_p(1 + n^{1/2}\|\theta - \theta_0\|) \tag{A5}$$

for any sequence $\delta_n \to 0$. This condition can be established by arguing exactly as in the proof of Theorem 1; the remaining details are therefore omitted. □

Proof of Theorem 3. Conditionally on the data and as $n \to \infty$,

$$\lim_{B\to\infty} \frac{1}{B}\mathfrak{Z}'\mathfrak{Z} = \begin{pmatrix} 1 & 0' \\ 0 & n^{-1}\hat{\Gamma}_G \end{pmatrix}, \quad \lim_{B\to\infty} \frac{1}{B}\mathfrak{Z}'\mathfrak{Y} = \begin{pmatrix} 0' \\ n^{-1}\hat{\Gamma}_G D_W \end{pmatrix} + o_p(n^{-1})$$

almost surely. Hence $\hat{D}_W = D_W + o_p(1)$, proving the first result. The second result follows from the first and the fact that $\hat{\Sigma}_W \to \Sigma_W$ in probability under Conditions A1–A10. □

Proof of Theorem 4. First, write $\hat{\Lambda}_0(t|\theta) - \Lambda_0(t) = \{\hat{\Lambda}_0(t|\theta) - \hat{\Lambda}_0(t|\theta_0)\} + \{\hat{\Lambda}_0(t|\theta_0) - \Lambda_0(t)\}$. Consider $\hat{\Lambda}_0(t|\theta) - \hat{\Lambda}_0(t|\theta_0)$. Under Conditions A1–A10, we may use Corollary A1 and arguments similar to those used in the proofs of Theorems 1 and 2 to show that

$$\hat{\Lambda}_0(t|\theta) - \hat{\Lambda}_0(t|\theta_0) \equiv \frac{1}{n} \sum_{i=1}^{n} \int_0^t \frac{Y_i(u|\theta)}{s^{(0)}(u|\theta)}[\lambda_0\{uh_i(\theta)\}h_i(\theta) - \lambda_0(u)]du + o_p(n^{-1/2}), \tag{A6}$$

where $h_i(\theta) = \exp\{(\theta_0 - \theta)'Z_i\}$ and the $o_p(n^{-1/2})$ term is uniform in t and $\theta \in \mathscr{N}(\theta_0)$. Supposing that $\|\theta - \theta_0\| \leqslant \delta_n$ for any $\delta_n \to 0$ and proceeding similarly to the proofs of Theorems 1 and 2, we can show that $\hat{\Lambda}_0(t|\theta) - \hat{\Lambda}_0(t|\theta_0) = -A_Q'(t)(\theta - \theta_0) + o_p(n^{-1/2})$, where $A_Q(t)$ is given in (10) and $\|\theta - \theta_0\| \leqslant n^{-1/2}M$. This establishes (23).

Turning to $\hat{\Lambda}_0(t|\theta_0) - \Lambda_0(t)$, we may use Corollary A1 to write

$$\hat{\Lambda}_0(t|\theta_0) - \Lambda_0(t) = n^{-1} \sum_{i=1}^{n} \tilde{\xi}_i(t|\theta_0) + o_p(n^{-1/2}),$$

where $\tilde{\xi}_i(t|\theta_0) = \int_0^t \{s^{(0)}(u|\theta)\}^{-1} M_i(du|\theta_0)$, the error being uniform for $t \in [0, \bar{\tau}]$. If we use (23) and the fact that $\hat{\theta}_W - \theta_0 = -D_W^{-1} \bar{S}_W(\theta_0) + o_p(n^{-1/2})$, it follows that $\hat{\Lambda}_0(t|\hat{\theta}_W) - \Lambda_0(t)$ is equivalent to $n^{-1} \sum_{i=1}^{n} \xi_i(t|\theta_0)$, where

$$\xi_i(t|\theta_0) = \tilde{\xi}_i(t|\theta_0) + A_Q'(t) D_W^{-1} \int_0^{\bar{\tau}} w(u|\theta_0) \left\{ Z_i - \frac{s^{(1)}(u|\theta_0)}{s^{(0)}(u|\theta_0)} \right\} M_i(du|\theta_0).$$

The generic summand $\xi(.|\theta_0)$ is a cadlag function of t and of bounded variation for $t \in [0, \bar{\tau}]$. Hence, $\xi(.|\theta_0)$ is $\mathbb{P}_0$-Donsker and $n^{-1/2} \sum_{i=1} \xi_i(t|\theta_0)$ converges weakly to a Gaussian process with covariance $E\{\xi_1(s|\theta_0)\xi_1(t|\theta_0)\}$ for any t,s in $[0, \bar{\tau}]$ (Parner, 1998, Lemma 2). Tedious but straightforward computations using Lemma 3 yield the covariance term; see also the proof of Proposition 2. The remaining details are omitted. □

References

Aalen, O. & Husebye, E. (1991). Statistical analysis of repeated events forming renewal processes. *Statist. Med.* **10**, 1227–40.

Aalen, O., Fosen, J., Weedon-Fekjaer, H., Borgan, O. & Husebye, E. (2004). Dynamic analysis of multivariate failure time data. *Biometrics* **60**, 764–73.

Andersen, P. K. & Gill, R. D. (1982). Cox's regression model for counting processes: a large sample study. *Ann. Statist.* **10**, 1100–20.

Andersen, P. K., Borgan, O., Gill, R. D. & Keiding, N. (1993). *Statistical Models Based on Counting Processes.* New York: Springer-Verlag.

Bickel, P. J., Klaassen, C. A. J., Ritov, Y. & Wellner, J. A. (1993). *Efficient and Adaptive Estimation for Semiparametric Models.* Baltimore: Johns Hopkins University Press.

Byar, D. (1980). The Veterans Administration study of chemoprophylaxis for recurrent stage I bladder tumors: comparisons of placebo, pyridoxine, and topical thiotepa. In *Bladder Tumours and Other Topics in Urological Oncology*, Ed. M. Pavone-Macaluso, P. H. Smith and F. Fedsmyn, pp. 363–70. New York: Plenum.

Chang, I. S. & Hsiung, C. A. (1994). Information and asymptotic efficiency in some generalized proportional hazards models for counting processes. *Ann. Statist.* **22**, 1275–98.

Chang, S. & Wang, M. C. (1999). Conditional regression analysis for recurrence time data. *J. Am. Statist. Assoc.* **94**, 1221–30.

Cox, D. (1972a). Regression models and life tables (with Discussion). *J. R. Statist. Soc.* B **34**, 187–220.

Cox, D. R. (1972b). The statistical analysis of dependencies in point processes. In *Stochastic Point Processes: Statistical Analysis, Theory, and Applications*, Ed. P. A. W. Lewis, pp. 55–66. New York: Wiley.

Daley, D. J. & Vere-Jones, D. (1988). *An Introduction to the Theory of Point Processes.* New York: Springer-Verlag.

Fleming, T. R. & Harrington, D. P. (1991). *Counting Processes and Survival Analysis.* New York: Wiley.

Fygenson, M. & Ritov, Y. (1994). Monotone estimating equations for censored data. *Ann. Statist.* **22**, 732–46.

Gill, R. D. (1980). Nonparametric estimation based on censored observations of a Markov renewal process. *Z. Wahr. verw. Geb.* **53**, 97–116.

Huang, Y. (2002a). Calibration regression of censored lifetime medical cost. *J. Am. Statist. Assoc.* **97**, 318–27.

Huang, Y. (2002b). Censored regression with the multistate accelerated sojourn times model. *J. R. Statist. Soc.* B **64**, 17–29.

Jin, Z., Lin, D. Y., Wei, L. J. & Ying, Z. (2003). Rank-based inference for the accelerated failure time model. *Biometrika* **90**, 341–53.

Jones, M. P. (1997). A class of semiparametric regressions for the accelerated failure time model. *Biometrika* **84**, 73–84.

Kalbfleisch, J. D. & Prentice, R. L. (2002). *The Statistical Analysis of Failure Time Data*, 2nd ed. New York: Wiley.

Lawless, J. & Nadeau, C. (1995). Some simple robust methods for the analysis of recurrent events. *Technometrics* **37**, 158–68.

Lin, D. Y., Wei, L. J., Yang, I. & Ying, Z. (2000). Semiparametric regression for the mean and rate functions of recurrent events. *J. R. Statist. Soc.* B **62**, 711–30.

LIN, D. Y., WEI, L. J. & YING, Z. (1998). Accelerated failure time models for counting processes. *Biometrika* **85**, 605–18.

LIN, D. Y., WEI, L. J. & YING, Z. (2001). Semiparametric transformation models for point processes. *J. Am. Statist. Assoc.* **96**, 620–8.

OAKES, D. (1992). Frailty models for multiple event times. In *Survival Analysis: State of the Art*, Ed. J. Klein and P. Goel, pp. 371–9. Dordrecht: Kluwer.

OAKES, D. & CUI, L. (1994). On semiparametric inference for modulated renewal processes. *Biometrika* **81**, 83–90.

PARNER, E. (1988). Asymptotic theory for the correlated gamma-frailty model. *Ann. Statist.* **26**, 183–214.

PEÑA, E. A., STRAWDERMAN, R. L. & HOLLANDER, M. (2001). Nonparametric estimation with recurrent event data. *J. Am. Statist. Assoc.* **96**, 1299–315.

PRENTICE, R. L. (1978). Linear rank tests with right censored data. *Biometrika* **65**, 167–79.

PRENTICE, R. L., WILLIAMS, B. & PETERSEN, A. (1981). On the regression analysis of multivariate failure time data. *Biometrika* **68**, 373–9.

REID, N. (1994). A conversation with Sir David Cox. *Statist. Sci.* **9**, 439–55.

RITOV, Y. (1990). Estimation in a linear regression model with censored data. *Ann. Statist.* **18**, 303–28.

RITOV, Y. (1991). Estimating functions in semi-parametric models. In *Estimating Functions*, Ed. V. Godambe, pp. 319–36. Oxford: Clarendon Press.

ROBINS, J. & TSIATIS, A. A. (1992). Semiparametric estimation of an accelerated failure time model with time-dependent covariates. *Biometrika* **79**, 311–9.

SCOTT, D. W. (1992). *Multivariate Density Estimation: Theory, Practice, and Visualization.* New York: Wiley.

TSIATIS, A. A. (1990). Estimating regression parameters using linear rank tests for censored data. *Ann. Statist.* **18**, 354–72.

VAN DER VAART, A. W. (1998). *Asymptotic Statistics.* Cambridge: Cambridge University Press.

VAN DER VAART, A. W. & WELLNER, J. A. (1996). *Weak Convergence and Empirical Processes.* New York: Springer-Verlag.

WANG, M. C. & CHANG, S. (1999). Nonparametric estimation of a recurrent survival function. *J. Am. Statist. Assoc.* **94**, 146–53.

YING, Z. (1993). A large sample study of rank estimation for censored regression data. *Ann. Statist.* **21**, 76–99.

[*Received September* 2003. *Revised January* 2005]